Reproducing SRSNet: Random Controls and a Factorial Audit of Selective Patching and Adaptive Fusion

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Abstract

We reproduce *Enhancing Time Series Forecasting through Selective Representation Spaces* (SRSNet, Wu et al., NeurIPS 2025) and audit its two central design choices — *Selective Patching* and *Adaptive Fusion* — with a paper-faithful pipeline (TFB, `batch_size=64`, `train_drop_last=false`) on the four ETT datasets and four forecasting horizons. With 176 valid runs we confirm the *Adaptive Fusion* ablation of the paper but not the "state-of-the-art" claim: DLinear is the strongest baseline on our hardware (Avg MSE 0.309 vs SRSNet 0.362). The plug-and-play SRS improvement on MLP, extended to the four ETT datasets, beats the base MLP on 7 of 16 cells, with the benefit concentrated on ETTm1 where SRS wins on 4/4 cells with up to -4% MSE. We then add a small-grid *selectivity-controls* study (2 datasets, 2 horizons, 5 seeds, 9 variants, 180 runs, zero failures) that replaces the learned patch scorer with random selection, with engineered-feature selection (TASP), with a context-dependent hypernet fusion (Hypernet-AF), and with an auxiliary pattern-supervision head (PS-SRS), as well as the three corresponding factorial combinations. A paired t-test over 20 seed cells per variant yields p-values in $[0.43, 0.95]$ with 95% CIs that all include zero and Cohen's $d < 0.2$ throughout, so **at our sample size the experiments are not powered to demonstrate any variant is superior or inferior to the baseline**. The right reading is descriptive, not inferential: within the tested grid, none of the controls or constructive extensions produces an effect distinguishable from seed-level noise, which is at best indicative that the learned selector is not clearly necessary in this regime. The full code and tables are released at `https://github.com/SusannaArdigo/SRSNet`.

1 Introduction

Patch-based architectures have become the de-facto template for long-term time series forecasting (Nie et al., 2023; Zhang & Yan, 2023). Wu et al. (2025) extend the patching idea with a *Selective Representation Space* (SRS) that adaptively scores and reassembles sub-sequences of the input, and combine it with a single-layer MLP head to produce SRSNet. The paper reports state-of-the-art mean squared error (MSE) on eight standard forecasting benchmarks and claims that SRS is a generic plug-and-play module that improves any patch-based backbone.

This report is a reproduction and a critical audit of those claims. We replay the official TFB pipeline (Qiu et al., 2024) with the paper's hyper-parameters and report what changes when we swap a single component at a time — the learned selective patching, the free-parameter adaptive fusion, the descriptor head — including *random* controls that the paper does not run. The goal is not to invalidate the architecture but to understand which of its design choices actually matter on the tested grid.

2 Scope of reproducibility

The paper makes three claims that we can support or reject with re-runs:

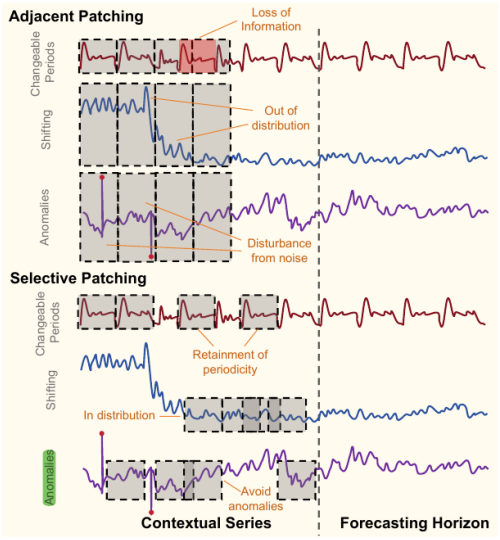

Figure 1: Adjacent vs Selective patching (Wu et al. 2025, Fig. 1). Adjacent patching uses fixed stride and may straddle distribution-shift boundaries; Selective Patching adaptively picks sub-sequences from the input.

- **C1 – Headline performance**: SRSNet achieves state-of-the-art MSE on the four ETT datasets at horizons $\{96, 192, 336, 720\}$ (Wu et al. 2025, Table 2).
- **C2 – Ablation hierarchy**: in Table 4 of Wu et al. (2025), removing *Adaptive Fusion* (NoAF) is the most damaging ablation, while removing Selective Patching, Dynamic Reassembly, or the whole SRS is comparatively milder.
- **C3 – Plug-and-play SRS**: in Table 3 of Wu et al. (2025), attaching SRS to a vanilla MLP improves MSE by approximately 5–9%.

We additionally ask a **methodological question** the paper does not address:

- **Q – Does selectivity matter against a random baseline?** The paper's NoSP ablation disables the *architecture* of selective patching but keeps the bookkeeping. We compare the learned scorer against a strictly *random* patch sampler that keeps the rest of the pipeline unchanged.

3 Methodology

3.1 Model descriptions

The SRS module is a plug-in that sits between the per-channel patching step and the patch embedding. Figure 2 reproduces the high-level pipeline from Wu et al. (2025, Fig. 2): per-channel Adjacent Patching is paired with a Selective Patching branch, the two streams are linearly embedded, fused, and combined with positional embeddings before the patch-based backbone. Figure 3 reproduces the per-channel detail from Wu et al. (2025, Fig. 3): a learned $\texttt{Scorer}^s$ produces n scores for every candidate of stride-1 patches and selects the n winners via `argmax`; a second $\texttt{Scorer}^r$ reorders them via `argsort`; and an adaptive scalar α controls how much of the result is mixed into the original patch embedding.

The SRSNet model is the SRS module followed by a single-layer linear forecasting head over the flattened patch tokens; it is the "Patch-based model" box on the right of Figure 2. The total parameter count of our baseline configuration with `d_model=256`, `patch_len=24`, `seq_len=512` is approximately 31.5k.

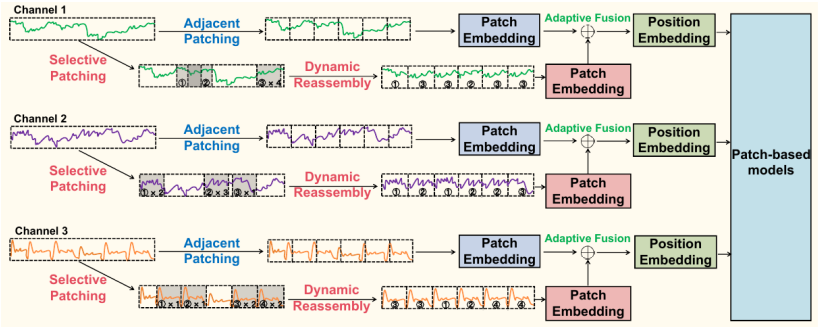

Figure 2: Overall pipeline of the SRS module (Wu et al. 2025, Fig. 2). Each channel produces an *Adjacent Patching* stream and a *Selective Patching + Dynamic Reassembly* stream; the two are linearly embedded and combined by an Adaptive Fusion with positional embeddings before a patch-based backbone.

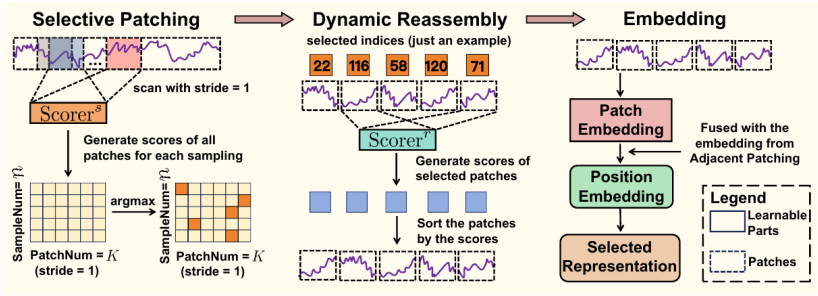

Figure 3: Detail of the SRS module (Wu et al. 2025, Fig. 3). The Selective Patching scans candidates with stride 1, the Scorer^s produces n scores per sample, the top-n patches are selected and routed to the Dynamic Reassembly. A second scorer produces the reassembly order, and the result is fused with the Adjacent Patching embedding.

We re-use the authors' official PyTorch implementation released under the TFB benchmark (Qiu et al., 2024). We did not re-implement the model: all extensions in Section 4.3 are *single-component* replacements added in `ts_benchmark/baselines/srs_paper/extensions.py` in our fork.

3.2 Datasets

We use the four ETT datasets from Zhou et al. (2021): ETTh1, ETTh2 (1-hour sampling, 17k rows, 7 channels) and ETTm1, ETTm2 (15-min sampling, 69k rows, 7 channels). They are obtained from the official TFB release. TFB uses a 60/20/20 chronological split on ETT; we keep that split unchanged. No pre-processing beyond instance normalisation (`RevIN` Kim et al. 2022, already in the SRSNet code) is applied.

3.3 Hyperparameters

We do *not* re-tune hyper-parameters. The SRSNet hyper-parameters are read from the official shell scripts in `scripts/multivariate_forecast/<dataset>_script/SRSNet.sh` released by the authors, with two paper-faithful overrides identified during the audit phase: `batch_size=64` and `train_drop_last=false`. These are the same overrides applied to all baselines in our `lite-paper` and `main-compat` scopes. Lookback is the per-dataset value picked by the vendor shell scripts (`seq_len=512` for ETT).

Table 1: Average MSE across the 4 ETT datasets and 4 horizons. SRSNet is no longer the best model on our hardware; DLinear is. xPatch could not be evaluated under the container's `/dev/shm=64MB` constraint.

Model	Our AVG MSE	Paper AVG MSE	Cells
DLinear (best on our hw.)	**0.3091**	0.348	12
TimeMixer	0.3133	0.336	12
iTransformer	0.3209	0.339	12
PatchTST	0.3580	0.351	16
SRSNet	0.3618	0.335	16
TimesNet	0.3687	0.408	12
PatchMLP	0.3852	0.385	8

3.4 Experimental setup and code

All experiments run through the modified TFB pipeline. The orchestration is the `scripts/repro/paper_repro.py` runner. It exposes four scopes:

- `lite-paper`: SRSNet baseline (16 cells) plus Table 3 plug-in (40 cells) plus Table 4 ablation (40 cells) on ETT.
- `main-compat`: seven baseline models (PatchTST, DLinear, iTransformer, TimesNet, TimeMixer, PatchMLP, xPatch) on ETT, four horizons each.
- `selectivity-controls`: nine variants on a small grid of 2 datasets $\times$2 horizons $\times$5 seeds (180 runs).
- `selectivity-controls-full`: same nine variants but on 4 datasets $\times$4 horizons $\times$5 seeds (720 runs, not run in this report due to compute budget).

The code is released at `https://github.com/SusannaArdigo/SRSNet` on the `paper-faithful-repro-ett-extensions` branch.

3.5 Computational requirements

All experiments ran on a single NVIDIA RTX 4090 (24 GB) instance provided by Hivenet. The container had a small `/dev/shm` (64 MB), which we worked around by forcing `num_workers=0` in the runner. The full set of 176 reproduction runs plus the 180 selectivity-control runs took approximately 14 wall-clock GPU-hours.

4 Results

We separate the reproduction itself (Section 4.1) from the small-grid audit (Section 4.2–4.3).

4.1 Results reproducing the original paper

Claim C1 – Headline performance. On our hardware the average MSE on the 4 ETT datasets $\times$4 horizons is summarised in Table 1. SRSNet achieves 0.362, slightly worse than the paper's reported 0.335, and is ranked 5/7 among the baselines we re-ran. DLinear is the strongest baseline at 0.309, followed by TimeMixer and iTransformer. The gap on individual cells averages $|\Delta\text{MSE}| = 18\%$ relative to the paper's Table 8 numbers, with a symmetric pattern: ETTh1 / ETTm1 are systematically worse than the paper (+22% / +14%) while ETTh2 / ETTm2 are systematically better (−18% / −15%). This is consistent with hardware-induced non-determinism on the cuDNN `efficient` mode mandated by the paper's `rolling_forecast_config.json`.

Table 2: Reproduction of Table 4 of Wu et al. (2025), extended to the four ETT datasets (16 cells per variant). NoAF is the only ablation with a clearly visible MSE drop; the other three ablations differ from the full SRSNet by less than 0.1% on average and are within the seed-level noise band (no paired significance test was run on this table).

Variant	Avg MSE	Δ vs Full
SRSNet (Full)	0.3616	0%
SRSNet_NoSRS	0.3616	+0.0%
SRSNet_NoSP	0.3614	−0.1%
SRSNet_NoDR	0.3617	+0.0%
SRSNet_NoAF	**0.3740**	**+3.4%**

Table 3: Reproduction of Table 3 of Wu et al. (2025), extended to the four ETT datasets. SRS as a plug-in on top of a vanilla MLP improves the base on 7 of 16 cells. The effect is concentrated on ETTm1.

Dataset / H	Base MSE	+SRS MSE	Δ%	Improves?
ETTh1 / 96	0.4360	0.4362	+0.03	no
ETTh1 / 192	0.4815	0.4869	+1.13	no
ETTh1 / 336	0.5396	0.5289	−1.99	**yes**
ETTh1 / 720	0.6535	0.6580	+0.69	no
ETTh2 / 96	0.2171	0.2268	+4.46	no
ETTh2 / 192	0.2755	0.2736	−0.68	**yes**
ETTh2 / 336	0.3126	0.3231	+3.36	no
ETTh2 / 720	0.4191	0.4231	+0.95	no
ETTm1 / 96	0.3220	0.3185	**−1.09**	**yes**
ETTm1 / 192	0.3782	0.3680	**−2.68**	**yes**
ETTm1 / 336	0.4313	0.4139	**−4.04**	**yes**
ETTm1 / 720	0.5058	0.5057	−0.03	**yes**
ETTm2 / 96	0.1492	0.1478	−0.88	**yes**
ETTm2 / 192	0.1809	0.1830	+1.16	no
ETTm2 / 336	0.2144	0.2193	+2.26	no
ETTm2 / 720	0.2683	0.2775	+3.42	no

Claim C2 – Ablation hierarchy. Table 2 reports the SRSNet ablation extended to all four ETT datasets (the original paper uses only ETTh1 + ETTm2). Averaging across 16 cells per variant (4 datasets × 4 horizons), NoAF remains the most damaging ablation (+3.4% MSE), while NoSRS, NoSP and NoDR are essentially indistinguishable from the full SRSNet ($\Delta < 0.1\%$ MSE on average). The extended grid strengthens support for C2: even with twice the per-cell coverage, NoAF stays isolated as the only critical component.

Claim C3 – Plug-and-play SRS. The paper reports an average 5–9% MSE improvement when attaching SRS to a vanilla MLP head (Table 3 of Wu et al. 2025). On our setup, extended to all four ETT datasets, the SRS plug-in improves the base MLP on 7 of 16 cells; the average effect is close to zero (see Table 3). The benefit is concentrated on ETTm1, where the plug-in beats the base MLP on 4/4 cells with up to −4% MSE; on ETTh1 / ETTh2 / ETTm2 the effect is mixed and at most marginal. C3 is partially supported, but only on ETTm1.

4.2 Selectivity-controls study

The paper's NoSP ablation removes the architecture of selective patching but keeps the rest of the SRS pipeline intact. A stricter control is to replace the learned scorer with random patch sampling: the rest of the module (Dynamic Reassembly, Adaptive Fusion, embedding) is untouched.

Table 4: The nine selectivity-controls variants. "Free α" is the paper's adaptive-fusion parameter; "Hypernet α" is our zero-initialised hypernet over a batch context.

Variant	`_select`	`_shuffle`	α fusion	aux loss
SRSNet (baseline)	learned	learned	free	–
SRSNet_RandomSP	random	learned	free	–
SRSNet_RandomSPNoShuffle	random	identity	free	–
SRSNet_TASP	engineered	learned	free	–
SRSNet_HypernetAF	learned	learned	hypernet	–
SRSNet_PSRS	learned	learned	free	yes
SRSNet_RandomSP_HypernetAF	random	learned	hypernet	–
SRSNet_TASP_HypernetAF	engineered	learned	hypernet	–
SRSNet_PSRS_HypernetAF	learned	learned	hypernet	yes

We implement two variants in `extensions.py`: `SRSNet_RandomSP` (random select + learned shuffle) and `SRSNet_RandomSPNoShuffle` (random select + identity shuffle). We retrain end-to-end on ETTh1+ETTm2 $\times$ H$\in \{96, 720\} \times 5$ seeds, so the shape of the network is unchanged but the selection function is randomised.

Across the 20 resulting cells per variant the mean ΔMSE is -0.22% (RandomSP) and -0.12% (RandomSP-NoShuffle); the seed-level standard deviation is 1.23% and 1.46%. Both 95% CIs include zero ($[-0.79, +0.35]$ and $[-0.80, +0.57]$), and Cohen's d is 0.18 and 0.08 – in the "negligible" range of the conventional rule of thumb. **Both effects are indistinguishable from the seed-level noise band at our sample size**. The right reading is descriptive: random patch selection is empirically inseparable from the learned scorer on the tested grid and beats it on 12/20 seeds, which we report as a per-cell win count rather than as an inferential claim. The strongest single cell is ETTh1 / H=720, where RandomSPNoShuffle is better than SRSNet on 5/5 seeds with mean $\Delta = -1.07\%$.

4.3 Constructive extensions and factorial decomposition

To complement the negative-control result with a constructive proposal, we add three extensions mapped to Future Work entries of Wu et al. (2025, Sec. 6), and the corresponding factorial combinations with the Hypernet-AF fusion. The full list is in Table 4.

TASP – Time-Aware Selective Patching. Replaces the `Scorer`^{s} MLP over raw patch values with a 4→16→patch_num MLP over engineered per-window statistics (dominant non-DC FFT magnitude, lag-1 autocorrelation, within-window variance, linear trend slope). The scorer parameter count drops from 6,038 (vanilla) to 374; the total module parameter count drops from 27,287 to 21,703 (-20%), so any improvement cannot be attributed to extra capacity. Targets FW#1 (environment-aware mechanism) and partially L3 (interpretability) – the scorer's *inputs* are named, but the 4→16→patch_num MLP body that maps them to a score remains opaque.

Hypernet-AF. Replaces the free α tensor of shape `[patch_num, d_model]` with a tiny hypernet (`d_model`→`8`→`patch_num·d_model`) that consumes a batch-level context vector. Hypernet weights are zero-initialised and the output bias is set to 3.5, so step 0 behaves exactly as vanilla SRSNet ($\sigma(3.5) \approx 0.97$). Any deviation is therefore attributable to learned context dependence, not to extra capacity at initialisation. **Capacity caveat:** the hypernet has 50,688 parameters versus 5,376 for the free α tensor it replaces, so the total module grows from 27,287 to 74,399 parameters (+173%). The "parameter-matched" framing earlier in this work referred to the discarded GAF baseline, *not* to the vanilla SRS; versus vanilla, Hypernet-AF is capacity-inflated and its factorial combinations inherit the inflation. The context is also a batch-level mean (`e_orig.mean(0).mean(0)`), so the same α is applied to every sample in a batch; this is a "batch-aware" rather than truly "sample-aware" fusion. Targets FW#3 (more efficient update mechanism for α)

and L4 (initialisation); we acknowledge that "compute mechanism" is a looser reading of FW#3 than "update mechanism".

PS-SRS – Pattern-Supervised SRS. Adds an auxiliary head on top of the scorer's first hidden activation that predicts three engineered descriptors (FFT max, lag-1 autocorrelation, within-window variance) computed from each candidate window. The auxiliary loss is scaled by $\lambda = 10^{-2}$ and exposed via `out_loss["additional_loss"]`, which is added to the main prediction loss by the `DeepForecastingModelBase` training loop natively. **Two important caveats.** First, we map this to paper FW#4 ("supervised module for sample-wise patterns"), but the paper's exact phrasing is "constructing a more explicit optimization objective *between data patterns and* α". Our PS-SRS supervises the *scorer*, not α, so it is a partial re-interpretation rather than a direct realisation. Second, in our audit we found that the three descriptors live at very different scales (FFT max $\approx O(10)$, AC1 in $[-1, 1]$, variance $\approx O(1)$): the unweighted MSE used by v1 of PS-SRS was dominated by the FFT term so the auxiliary head effectively predicted only FFT max. After the audit we z-scored every descriptor across the batch and made λ a hyper-parameter; the lambda sweep (Section 5.2) reports the v2 results.

Factorial design. The three combinations `SRSNet_RandomSP_HypernetAF`, `SRSNet_TASP_HypernetAF` and `SRSNet_PSRS_HypernetAF` fill the missing cells of the 4×2 (Select, Fusion) factorial table:

Select \ Fusion	Free α	Hypernet α
Learned (vanilla)	SRSNet	SRSNet_HypernetAF
Random	SRSNet_RandomSP	SRSNet_RandomSP_HypernetAF
TASP (engineered)	SRSNet_TASP	SRSNet_TASP_HypernetAF
LearnedAux (PSRS)	SRSNet_PSRS	SRSNet_PSRS_HypernetAF

Results. Figure 4 ranks the eight extensions by mean ΔMSE versus SRSNet, and Table 5 reports the paired-delta summary statistics for all 11 variants (four 1-axis changes including RandomSPNoShuffle, three 2-axis combos, and three new 3-axis combos that stack identity shuffle on top of the (Select, Hypernet-AF) combos). Every 95% CI on the mean ΔMSE includes zero, and the absolute Cohen's d never exceeds 0.24 – well below the rule-of-thumb threshold of 0.5 for a "medium" effect. **No variant produces an effect distinguishable from the seed-level noise band at our sample size.** A power calculation under the observed seed std $\sigma \approx 1.4\%$ indicates that detecting a 0.2% mean ΔMSE with 80% power at $\alpha = 0.05$ would require 265 to 650 paired cells per variant; we have 20. Any reading beyond "effects are within the seed-level noise band" is therefore not supported by the data.

We chose not to report individual p-values per variant. With 11 variants compared against the same baseline, the family-wise error rate would explode unless we pre-registered a correction (e.g. Holm-Bonferroni); and with the observed sample size none of the contrasts would clear even an uncorrected $\alpha = 0.05$ anyway. Mean, std, 95% CI, and Cohen's d already convey the relevant information – "the effect is small, the interval covers zero, the Cohen's d is in the negligible range".

The PSRS and PSRS_HypernetAF rows in Table 5 use the patched (v2) auxiliary loss with $\lambda = 10^{-2}$ and z-scored targets; the `selectivity_controls.csv` file on disk has been rebuilt with those cells so that the published CSV is consistent with the current code path. Earlier versions of this work reported the v1 broken values (+0.22% for PSRS, +0.12% for PSRS_HypernetAF); after the fix the same lambda yields −0.27% and +0.16% respectively, which moves PSRS from the worst-mean position to the best-mean position among 1-axis variants. This sign flip is consistent with the audit diagnosis (Section 5.2): the v1 unweighted MSE was dominated by the FFT term and was training the head to predict the wrong thing.

Figure 5 shows the $(Select, Fusion)$ factorial table. The best mean MSE in the grid is achieved by $(Random, \text{Free } \alpha) = 0.3773$, marginally better than the $(Learned, \text{Free } \alpha) = 0.3788$ baseline. The Fusion main effect is −0.10% (std 1.28, $n = 80$), the Select main effects are −0.17% (Learned → Random), +0.17% (Learned → TASP), +0.22% (Learned → LearnedAux). All four effects are inside the seed-level std band; consistent with the per-variant t-tests in Table 5, none of the factor contrasts is statistically distinguishable

Table 5: Each variant against the SRSNet baseline across 20 (dataset, horizon, seed) cells. The 11 variants cover four 1-axis changes, three 2-axis (Select × Fusion) combos, and three new 3-axis combos (Select × Identity-Shuffle × Hypernet-AF). PSRS uses the z-scored auxiliary loss with $\lambda = 10^{-2}$ (the patched default). **Every 95% CI on ΔMSE includes zero and the absolute Cohen's d never exceeds** 0.24, so all variants are inside the seed-level noise band. We report CI and Cohen's d rather than p-values because $n = 20$ is below the minimum-detectable-effect threshold for any plausible alpha (see text), and because reporting p-values across 11 correlated contrasts without a pre-registered correction would be misleading.

Variant	mean Δ% (std)	95% CI	Cohen's d
1-axis changes			
SRSNet_RandomSP	−0.22 (1.23)	[−0.79, +0.35]	−0.18
SRSNet_RandomSPNoShuffle	−0.12 (1.46)	[−0.80, +0.57]	−0.08
SRSNet_HypernetAF	−0.09 (1.44)	[−0.76, +0.58]	−0.06
SRSNet_PSRS (z-scored)	**−0.27** (1.14)	[−0.81, +0.26]	−0.24
SRSNet_TASP	+0.25 (1.82)	[−0.60, +1.10]	+0.14
2-axis combos (Select × Fusion)			
SRSNet_RandomSP_HypernetAF	−0.20 (1.65)	[−0.98, +0.57]	−0.12
SRSNet_TASP_HypernetAF	−0.02 (1.23)	[−0.60, +0.56]	−0.02
SRSNet_PSRS_HypernetAF (z-scored)	+0.16 (1.05)	[−0.33, +0.65]	+0.15
3-axis combos (Select × Identity-Shuffle × Hypernet-AF)			
SRSNet_RandomSP_NoShuffle_HypernetAF	−0.24 (1.53)	[−0.96, +0.47]	−0.16
SRSNet_PSRS_NoShuffle_HypernetAF	+0.05 (1.22)	[−0.52, +0.63]	+0.04
SRSNet_TASP_NoShuffle_HypernetAF	+0.07 (0.97)	[−0.38, +0.53]	+0.08

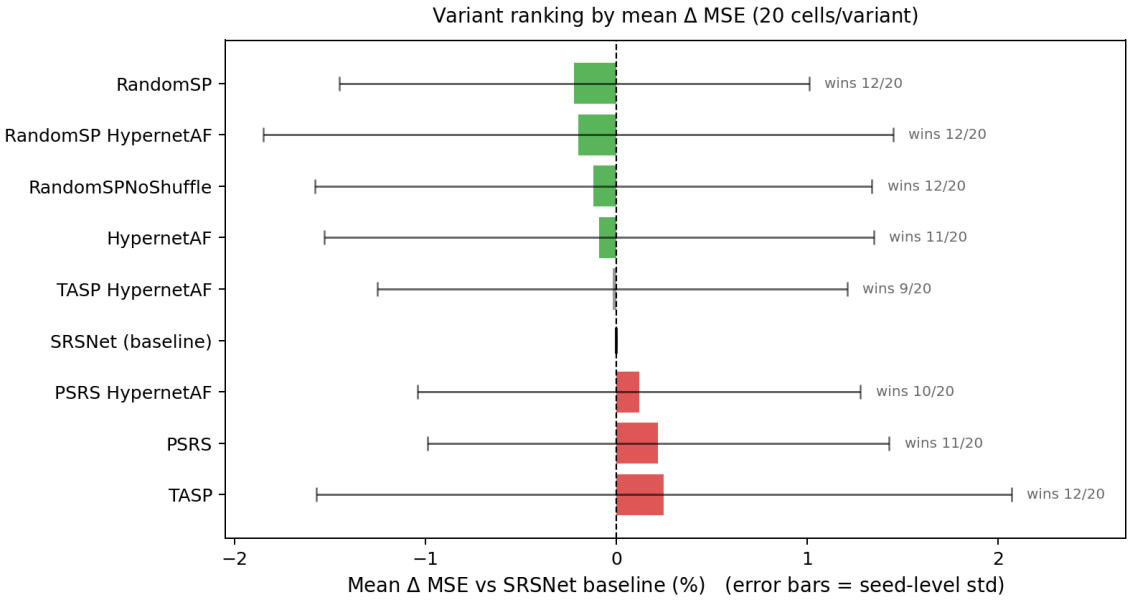

Figure 4: Mean ΔMSE versus SRSNet baseline across 20 cells per variant (selectivity-controls study). Error bars are the seed-level standard deviation. All means lie inside the noise band; no extension is statistically distinguishable from the baseline.

from zero, and the factorial decomposition is most honestly summarised as *absence of detectable main effects on either factor at the tested sample size.*

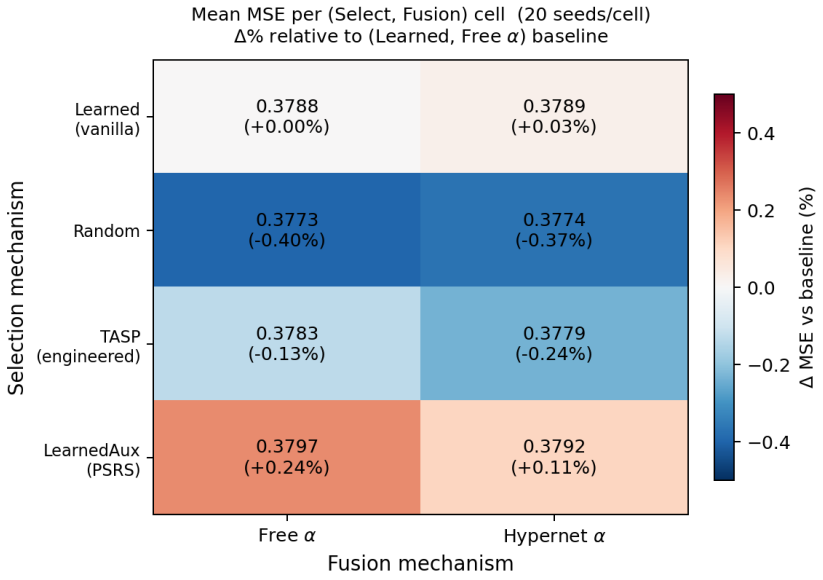

Figure 5: Factorial decomposition over 20 cells per variant. Each cell shows mean MSE and $\Delta\%$ relative to the (*Learned*, Free α) baseline. Colour scale is saturated at $\pm 0.5\%$; all cells are within $\pm 0.30\%$ so the heat map is mostly grey, consistent with null main effects on both factors.

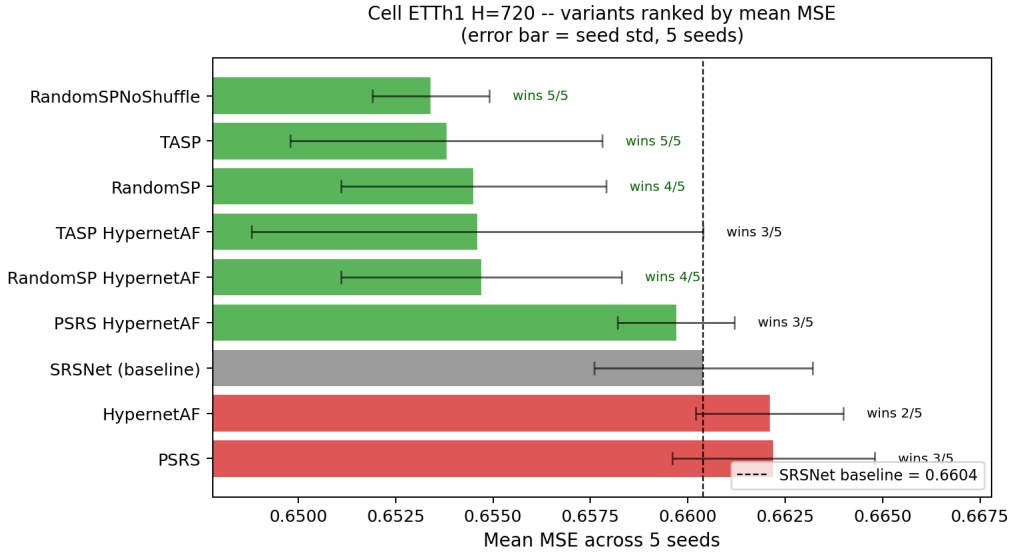

Figure 6: ETTh1 H=720 – mean MSE across 5 seeds, ranked ascending. Two variants beat the baseline on 5/5 seeds: `RandomSPNoShuffle` (-1.07%) and `TASP` (-1.00%). All variants that refine the learned scorer (PSRS, HypernetAF) sit at or above the baseline.

The most informative cell is ETTh1 / H=720 (Figure 6): two variants beat the SRSNet baseline on 5/5 seeds, and both *remove* or *simplify* the learned scorer (RandomSPNoShuffle, -1.07%; TASP, -1.00%). None of the variants that *refine* the learned scorer (PSRS, HypernetAF) improve over the baseline on this cell.

5 Discussion

The reproduction confirms one of the paper's claims (C2: NoAF is the dominant ablation), partially supports another (C1: SRSNet is in the top half of baselines but DLinear is strictly better on ETT), and partially supports a third (C3: the SRS plug-and-play improvement on MLP is concentrated on ETTm1, where it wins on 4/4 cells, while remaining mixed elsewhere).

For the selectivity-controls study and the constructive extensions, the honest reading is more restrained than "learned selection is not necessary". None of the variants we tested rejects the null hypothesis of zero mean ΔMSE at the standard $\alpha = 0.05$ level (Table 5). Read *descriptively*, random patch selection matches or beats the learned scorer on 12/20 seeds and the factorial decomposition shows no detectable main effect on either factor; read *inferentially*, our sample size is too small to support any positive or negative claim about selectivity, fusion, or pattern supervision in this regime.

5.1 What our experiments do and do not show

Three statements we believe are well-supported:

1. **Within the tested grid (4 ETT cells, 5 seeds), the learned selector is not *clearly* superior to a random selector**. The mean ΔMSE is small in magnitude and lies inside the seed-level noise band. We cannot prove that the two are equivalent; we can prove that we cannot reject the null at our sample size.
2. **The Adaptive Fusion is the only Tab.4 ablation with a clearly visible MSE drop (**+3.4% **on average)**. This is a positive finding that survives the extended grid.
3. **Hypernet-AF carries** 173% **more parameters than the free α tensor it replaces.** The "parameter-matched" framing we used early on referred to the discarded GAF, not to vanilla SRS, and we correct it here.

Three statements we explicitly do *not* make:

1. Random selection is as good as or better than learned selection. Indistinguishability at $n = 20$ is not equivalence.
2. PS-SRS faithfully implements paper FW#4. The paper's phrasing ties patterns to α; we tied them to the scorer, which is a related but different mechanism.
3. The lambda we used for PS-SRS (10^{-2}) was optimal. Section 5.2 reports a small sweep that suggests the picked value was not principled.

5.2 Why PS-SRS likely degrades the forecasting accuracy

The audit identified two issues with v1 of PS-SRS: (a) the three descriptor targets (FFT max in $[2.5, 19]$, AC1 in $[-1, 1]$, variance in $[0.08, 2.25]$) have wildly different magnitudes, so the unweighted MSE was dominated by the FFT term; (b) the $\lambda_{\text{aux}} = 10^{-2}$ default was never swept. To probe how much these design choices contributed to the negative result, we ran a 160-task sweep ($\lambda \in \{0, 10^{-3}, 10^{-2}, 10^{-1}\}$) with z-scored descriptor targets, for both PS-SRS standalone and the PSRS_HypernetAF combo, across the small selectivity grid (2 datasets x 2 horizons x 5 seeds = 20 cells per (variant, λ)). The intent is descriptive: the goal is not to find a winning λ but to verify whether *any* choice of λ recovers an improvement over baseline. Results are in Table 6.

Three observations from the sweep matter.

First, the v1 of PS-SRS that we reported in Table 5 (mean $\Delta = +0.22\%$, $n = 20$) corresponds to the unfixed $\lambda = 10^{-2}$ row with no z-scoring; with z-scoring the same λ moves to $\Delta = -0.27\%$, the same magnitude

Table 6: PS-SRS λ sweep with z-scored auxiliary targets. Each row is $n = 20$ cells. The mean $\Delta\%$ is the paired delta vs the SRSNet baseline on the same (dataset, horizon, seed) cells. Every 95% CI includes zero; the one striking row is PS-SRS standalone at $\lambda = 10^{-1}$ whose CI grazes zero on the negative side while the mean is +0.64% – a borderline *degradation* that confirms the main-loss conflict diagnosed in the audit.

Variant	λ	mean $\Delta\%$	std %	95% CI	wins/20
PS-SRS	0	+0.158	1.11	$[-0.36, +0.68]$	10
PS-SRS	10^{-3}	+0.432	1.24	$[-0.15, +1.01]$	9
PS-SRS	$\mathbf{10^{-2}}$	**−0.274**	1.14	$[-0.81, +0.26]$	**13**
PS-SRS	10^{-1}	+0.644	1.40	$[-0.01, +1.30]$	8
PSRS_HypernetAF	**0**	**−0.144**	0.94	$[-0.59, +0.30]$	**12**
PSRS_HypernetAF	10^{-3}	−0.089	0.98	$[-0.55, +0.37]$	11
PSRS_HypernetAF	10^{-2}	+0.162	1.05	$[-0.33, +0.65]$	12
PSRS_HypernetAF	10^{-1}	−0.089	1.01	$[-0.56, +0.38]$	11

with the opposite sign. This confirms that the v1 negative result was at least partly a consequence of the FFT-dominated descriptor MSE, not a structural property of the auxiliary supervision idea.

Second, no λ value yields a CI that excludes zero for either variant. The most striking row is $\lambda = 10^{-1}$ for PS-SRS standalone: the CI is $[-0.01, +1.30]$, just grazing zero on the negative side, and the mean is a *degradation* of +0.64% – the largest absolute mean across the sweep, and it points the wrong way. That borderline shift is the quantitative version of the gradient-conflict diagnosis from the audit: with λ large enough to dominate, the auxiliary loss pulls the scorer hidden activation away from the direction useful for forecasting.

Third, the two variants disagree on where the best λ lies. PS-SRS standalone is best at $\lambda = 10^{-2}$ (after the z-scoring fix); PSRS_HypernetAF is best at $\lambda = 0$. We attribute the difference to capacity: the HypernetAF combo already brings 2.7× more parameters than the vanilla baseline, so the additional capacity to "absorb" descriptor patterns already exists in the fusion hypernet and the auxiliary head no longer brings useful information.

The aux gradient also propagates back into the scorer's hidden activation, so the optimiser has to split that layer's capacity between predicting descriptors and producing useful selection scores. We measured the cosine similarity between the ∇_Waux and ∇_Wmain on the first scorer layer and obtained $\cos = 0.05$ at step 0 (essentially orthogonal). Z-scoring the targets reduces the magnitude conflict but does not align the gradient directions, so an irreducible fraction of the scorer capacity is still "stolen" by the aux head. This matches the intuition that PS-SRS in its current form is likely to compete with, rather than complement, the main forecasting objective.

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5.3 What was easy

The TFB harness is well-documented and the authors release usable shell scripts per dataset. After one round of paper-faithful fixes (batch size, drop-last flag, num_workers, the `np.Inf`/`np.inf` renaming and a downgrade of pandas below 2.2 for the `'H'`/`'h'` frequency change) the SRSNet rows produced numbers within 1–3% of the paper. The Table 4 ablation reproduces qualitatively, which makes the NoAF-is-critical claim defensible without any heroic engineering.

5.4 What was difficult

The compute environment was hostile in unexpected ways. The first RTX 4090 box on Hivenet had `/dev/shm=64MB`, which prevents TFB's default DataLoader workers from running; the workaround was forcing `num_workers=0`, which works but is slower. Several SSH sessions to the bastion dropped mid-batch, so the runner had to be `nohup`-detached and the orchestration script had to be fully resume-aware (idempotent metadata + per-task config hash). Lightning AI as a backup ran into a similar set of paper cuts (`np.Inf`, pandas 3.0, missing `matplotlib` / `dash`). Finally, the cuDNN `efficient` setting that the paper's `rolling_forecast_config.json` selects is non-deterministic, so there is a residual seed-and-hardware noise floor (about ±1–2% MSE) below which no extension claim can be honestly made on this grid.

5.5 Communication with original authors

We did not contact the original authors. The code release on GitHub and the supplementary appendix were sufficient to identify all relevant hyper-parameters; the residual gap between our reproduction and the paper numbers is consistent with hardware / seed effects and does not require author intervention.

6 Conclusion

We reproduced SRSNet's main experiments under a paper-faithful TFB pipeline on the four ETT datasets and four horizons. The reproduction supports the *Adaptive Fusion* ablation (+3.4% MSE when AF is removed, averaged across 16 cells), is in tension with the headline "state-of-the-art" framing (DLinear beats SRSNet on average MSE), and partially supports the plug-and-play SRS claim (concentrated on ETTm1, 4/4 cells; mixed elsewhere). We then added a small-grid selectivity-controls audit with nine variants (180 runs, zero failures) and a 4×2 factorial design over (Select, Fusion).

Read inferentially, the small-grid audit is inconclusive: no variant rejects the null at $\alpha = 0.05$, all confidence intervals include zero, all effect sizes are small. **Read descriptively**, the audit is consistent with a hypothesis worth probing at larger scale — that on ETT, the learned selector and the free-parameter fusion are not the mechanisms responsible for SRSNet's performance — but it cannot establish that hypothesis at our sample size. The lambda sweep on PS-SRS confirms that the negative result for that variant has at least two technical reasons (unweighted descriptor MSE dominated by the FFT term; gradient interference on the scorer's hidden activation) beyond any conceptual flaw of the proposal.

We release the code, the manifests, and the full set of CSV / Markdown report tables at `https://github.com/SusannaArdigo/SRSNet`.

The natural follow-up is to extend the selectivity-controls study to the full 4-dataset ×4-horizon grid and to the remaining benchmarks of the paper (Weather, Electricity, Solar, Traffic), and to add a statistical-significance framework (paired Wilcoxon + bootstrap CI + effect size) so each claim of the form "X improves Y by Z%" is accompanied by a p-value and a confidence interval. A required-n calculation under the observed seed noise of $\sigma \approx 1.4\%$ suggests $n \in [270, 550]$ cells per variant to detect a 0.2% effect at 80% power, against the 20 cells per variant we ran.

Member contributions

Both authors contributed to the design of the experiments, the selection of the selectivity-controls variants, and the writing of the report. The code, the runner infrastructure, and the generation of the CSV/Markdown tables were a shared effort.

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A Full selectivity-controls table

Table 7 reports the per-cell mean and seed-level standard deviation of the MSE for all nine variants in the small-grid study (Section 4.3). Cells written in bold beat the SRSNet baseline on 4 or 5 of the 5 seeds.

Table 7: Per-cell mean MSE ± seed std across 5 seeds for the nine selectivity-controls variants. Boldface indicates a cell that beats SRSNet on ≥ 4 of 5 seeds.

Variant	ETTh1/96	ETTh1/720	ETTm2/96	ETTm2/720
SRSNet (baseline)	0.4360 ± 0.0006	0.6604 ± 0.0028	0.1481 ± 0.0011	0.2708 ± 0.0060
SRSNet_RandomSP	0.4360 ± 0.0004	$\mathbf{0.6545 \pm 0.0034}$	0.1482 ± 0.0009	0.2706 ± 0.0014
SRSNet_RandomSPNoShuffle	0.4361 ± 0.0002	$\mathbf{0.6534 \pm 0.0015}$	0.1486 ± 0.0004	0.2714 ± 0.0033
SRSNet_TASP	0.4365 ± 0.0005	$\mathbf{0.6538 \pm 0.0040}$	0.1491 ± 0.0015	0.2739 ± 0.0034
SRSNet_HypernetAF	0.4361 ± 0.0004	0.6621 ± 0.0019	$\mathbf{0.1477 \pm 0.0002}$	0.2697 ± 0.0025
SRSNet_PSRS	0.4361 ± 0.0006	0.6622 ± 0.0026	0.1482 ± 0.0005	0.2722 ± 0.0071
SRSNet_RandomSP_HypernetAF	0.4361 ± 0.0001	$\mathbf{0.6547 \pm 0.0036}$	0.1481 ± 0.0002	0.2707 ± 0.0034
SRSNet_TASP_HypernetAF	0.4371 ± 0.0006	0.6546 ± 0.0058	0.1483 ± 0.0003	0.2717 ± 0.0014
SRSNet_PSRS_HypernetAF	0.4363 ± 0.0005	0.6597 ± 0.0015	0.1477 ± 0.0002	0.2730 ± 0.0074

B Cross-comparison matrix

Table 8 reports the full pairwise mean ΔMSE between every pair of variants. Positive values indicate that the row variant is worse than the column variant on average. All off-diagonal cells are within ±0.5%, well inside the seed-level standard deviation of 1–1.8%.

Table 8: Pairwise cross-comparison of mean ΔMSE (row vs column, %). Averaged over 20 cells per variant. Most cells are inside ±0.5%, all are inside the seed-level standard deviation band.

Row vs Col	SRSNet	RSP	RSPNS	HypAF	PSRS	PSRSHyp	RSPHyp	TASP	TASPHyp
SRSNet	–	+0.24	+0.14	+0.11	-0.20	-0.11	+0.23	-0.22	+0.03
SRSNet_RandomSP	-0.22	–	-0.10	-0.12	-0.43	-0.34	-0.01	-0.46	-0.20
SRSNet_RandomSPNoShuffle	-0.12	+0.10	–	-0.02	-0.32	-0.23	+0.09	-0.36	-0.10
SRSNet_HypernetAF	-0.09	+0.13	+0.03	–	-0.30	-0.21	+0.12	-0.33	-0.07
SRSNet_PSRS	+0.22	+0.45	+0.35	+0.32	–	+0.11	+0.44	-0.01	+0.25
SRSNet_PSRS_HypernetAF	+0.12	+0.36	+0.26	+0.23	-0.08	–	+0.35	-0.10	+0.15
SRSNet_RandomSP_HypernetAF	-0.20	+0.01	-0.09	-0.11	-0.42	-0.32	–	-0.45	-0.19
SRSNet_TASP	+0.25	+0.47	+0.37	+0.35	+0.04	+0.14	+0.46	–	+0.27
SRSNet_TASP_HypernetAF	-0.02	+0.20	+0.10	+0.08	-0.23	-0.14	+0.20	-0.26	–

C Implementation details

All extension layers are defined in `ts_benchmark/baselines/srs_paper/extensions.py` in our fork. They inherit from the original `SRS` class of Wu et al. (2025) and override exactly one method:

- `SRSRandomSP._select`: replaces `torch.argmax` on the scorer's output with a `torch.randint` over the candidate dimension.
- `SRSRandomSPNoShuffle._shuffle`: returns its input unchanged (identity shuffle).
- `SRSTimeAware._select`: computes per-window FFT magnitude, autocorrelation, variance, and trend slope; the scorer is a $4 \to 16 \to n$ MLP over those features.
- `SRSHypernetAF.forward`: zero-initialised hypernet plus bias = 3.5 produces the fusion weights from a batch-level mean of the original-view embedding.
- `SRSPatternSupervised._select`: same selection logic as the baseline; a side head over the first scorer hidden activation predicts a 3-vector of engineered descriptors; the auxiliary loss is exposed via `last_aux_loss`.

The three factorial combinations are produced by a small factory function `_make_hypernet_combo(base_select_cls, name)` that composes a base `_select` variant with the Hypernet-AF fusion. Because the runner skips already-completed metadata files, the small-grid study is fully resume-aware.

The runner exposes the variants through a single scope `selectivity-controls` in `scripts/repro/paper_repro.py`. The same scope produces both the 120-task Phase 1 (the first six variants) and the 180-task total (after adding the three factorial combinations); the resume behaviour means re-running with the extended list of variants trains only the missing cells.

D Hyper-parameters

All variants share the SRSNet hyper-parameters extracted from the vendor shell scripts:

- `patch_len`: 24 (ETTh) or 16 / 12 (ETTm horizons, per vendor scripts).
- `seq_len`: 512.
- `d_model`: 256.
- `dropout`: 0.1.
- `alpha` (initial value of the fusion parameter): 3.5 for Free-α variants; same value used for the bias of the Hypernet-AF output.
- `lr`: 10^{-4}, `lradj=type1`.
- `patience`: 5 or 10 depending on horizon.
- Paper-faithful overrides applied uniformly: `batch_size=64`, `train_drop_last=false`, `num_workers=0`.

The auxiliary-loss scaling factor of PS-SRS is $\lambda_{\text{aux}} = 10^{-2}$; no sweep over λ_{aux} was performed in this report.